

Exploratory Data Analysis

1. What does the name mean?

Exploratory comes from the word explore. To explore is to look around something new so you can learn what is there, like walking into a new town and seeing the streets, shops and people before you make any plan.

Data means information: numbers, names, dates or measurements. For example, how much rain fell each month, or how many kilograms of beans a farm produced.

Analysis means looking at something carefully to understand it and to find useful information.

In one sentence: Exploratory Data Analysis (EDA) means looking carefully at your data to learn what is inside it, before you do anything big with it. It is like checking the soil before you plant.

2. Why is EDA important?

If you skip EDA, you may decide based on data you do not really understand. EDA helps you to:

- **Find mistakes:** typing errors, extra zeros, wrong units.
- **Spot unusual values:** for example, one farm that is very far from all the others.
- **See patterns:** for example, more rain often goes with a bigger harvest.
- **Decide the next step:** you can choose what to do with the data with more confidence.

3. The questions EDA asks

EDA does not need advanced mathematics. It starts with simple questions:

1. How many rows (records) do I have?
2. What are the smallest and the biggest values?
3. What is the average?
4. Is anything missing or strange?
5. Do some things seem to go together?

4. Types of EDA

EDA can be divided in two simple ways. The first is by how many things you look at the same time:

- **Univariate:** one thing only. Example: the bean harvest alone (smallest, biggest, average).
- **Multivariate:** three or more things together. Example: rain, soil type and harvest.

The second is by how you look at them:

- **Non-graphical:** using numbers only, such as smallest, biggest, average, counts and tables.
- **Graphical:** using pictures, such as bar charts, line charts, scatter plots and histograms.

Together: you can mix them. A bar chart of harvests from each farm is univariate and graphical. A scatter plot of rain against harvest is bivariate and graphical. Some books call any case with two or more things “multivariate”, so the names can differ, but the idea is the same.

5. A small example

Here are records from five farms. Two things look wrong at first sight: Farm D reports 5000 kg, and Farm C has no rain value.

Farm	Beans (kg)	Rain (mm)
A	120	80
B	135	95
C	110	missing
D	5000	90
E	125	85

Outlier. Four farms harvested between 110 and 135 kg, but Farm D says 5000 kg. This far-away value is called an outlier. It could be a typing mistake (an extra zero) or a real, very large farm. If we do not check, the average becomes 1,098 kg, which does not describe any farm. The rule is: never delete a strange value without asking why.

Missing value. Farm C has an empty rain cell. We can ask for the real value, fill it in carefully (for example with the average), or remove that row if needed.

6. The main steps of EDA

1. **Look** at the data: see the rows and columns.
2. **Clean** the data: fix mistakes and handle missing values.
3. **Summarize** it: find the smallest, biggest and average values.
4. **Draw** simple charts to see the data quickly.
5. **Learn** from it: spot patterns and decide what to do next.

These steps are not strict. In real work you often go back and repeat them as you learn more.

7. Simple charts that help

- **Bar chart:** compares groups, such as which farm harvested the most.
- **Line chart:** shows change over time, such as rain in each month.
- **Scatter plot:** shows if two things go together, such as rain and harvest. If the dots go up to the right, more of one goes with more of the other.

Be careful: two things going together does not always mean that one causes the other. A pattern found in EDA is a clue, not final proof.

8. Summary

EDA is the first and most important look at your data. You explore the data, find the smallest, biggest and unusual values, check for missing or wrong entries, and use simple charts to see patterns. Only after that do you decide what to do next.

Look first. Understand. Then decide.

Sources

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Note: the farm numbers (beans and rain) in this summary are made-up examples for illustration only, not real data.